

CIVL4340

Wind Engineering

Week 3: Wind climate analysis continued

Overview

- **Wind climate analysis**
 - Sub- or super-set analysis
 - Directional multipliers
 - Division by storm type
 - Superstations
 - Data compositing
 - Data corrections

Reading

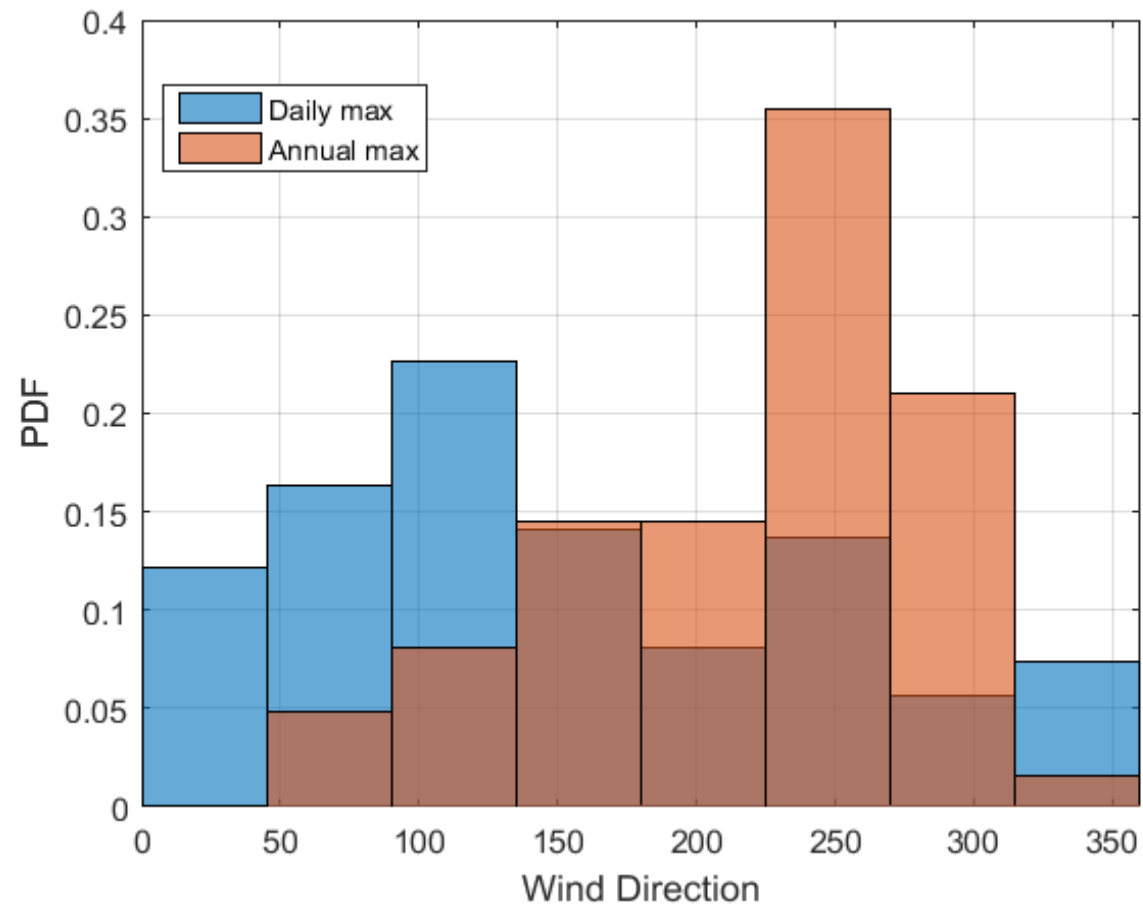
- Holmes, Chapter 2 & 3
- Simiu and Yeo, Chapter 3

Wind direction

- Structural response is sensitive to wind direction.
- Winds are equally likely from all directions.
 - Is this universally true? **No**
 - Storm type plays a role in determining directional probability.
- Some regions will have a relationship between directional distribution of the parent and of extremes.
 - Synoptic event extrema - **TRUE**
 - Localised/rare event extrema - **MAYBE**

Wind direction

- What does Brisbane look like?



Wind direction

- Directional analysis for subtropical regions (e.g. Brisbane), must be done using extreme data of interest.
- Approach: divide data into direction bins and calculate as before!

$P(\text{no events} > \text{given wind speed})$

$P(\text{no events} > \text{given wind speed from a given dirn, } \theta_i)$

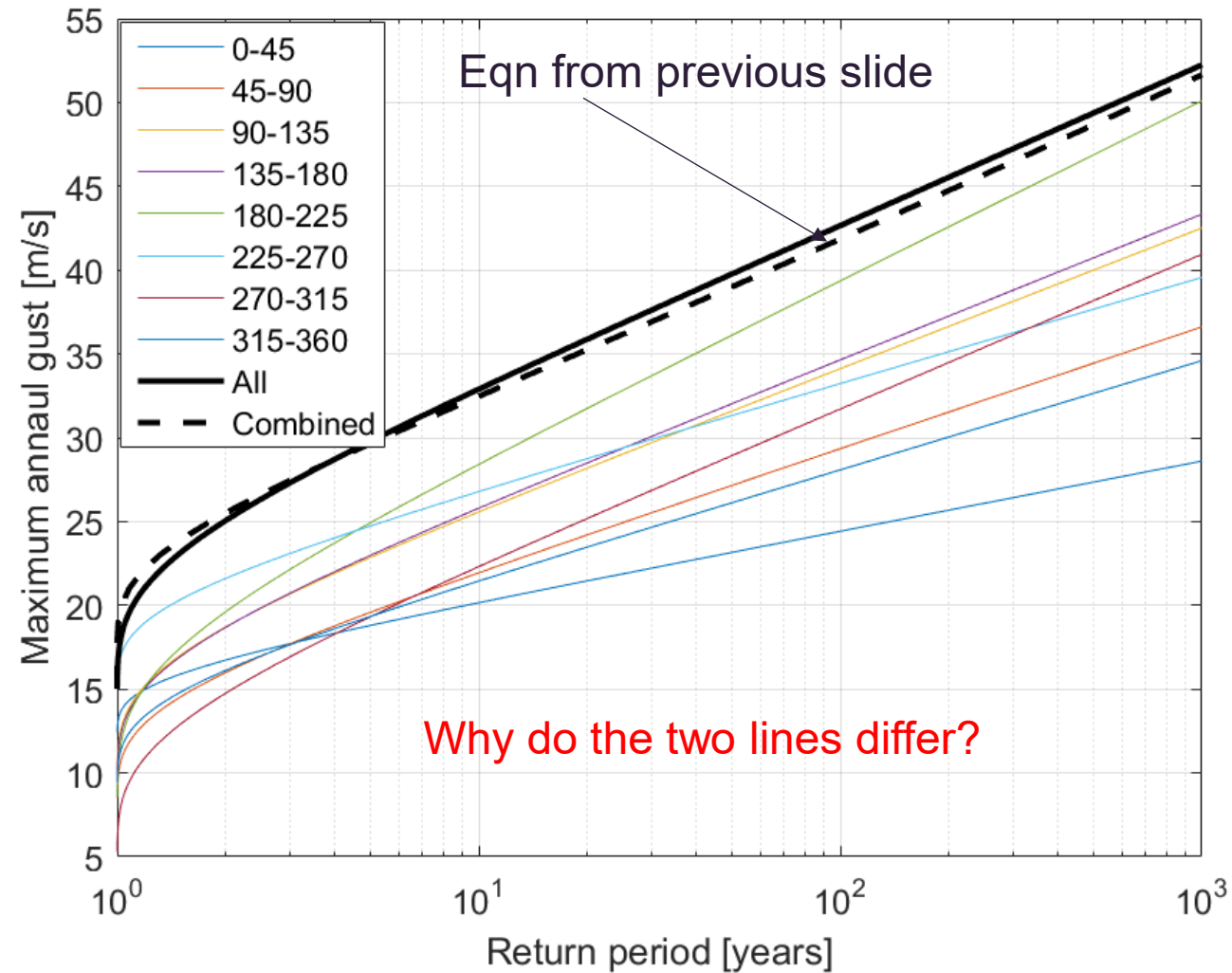
$$P(X_{all} = 0) = \prod_{i=1}^N P(X_{\theta_i} = 0)$$

$$\left(1 - \frac{1}{R_{all}}\right) = \prod_{i=1}^N \left(1 - \frac{1}{R_{\theta_i}}\right)$$

Return period for gust with data from all directions

Return period for gust from a given direction, θ_i

Wind direction



Wind direction

- Directional multipliers often used in design codes/standards (this is the case for AS/NZS1170.2).
- Impractical to have different return period curves for each direction – simplified approaches developed.

Methods – see *sect. 2.2.7 in Holmes text*.

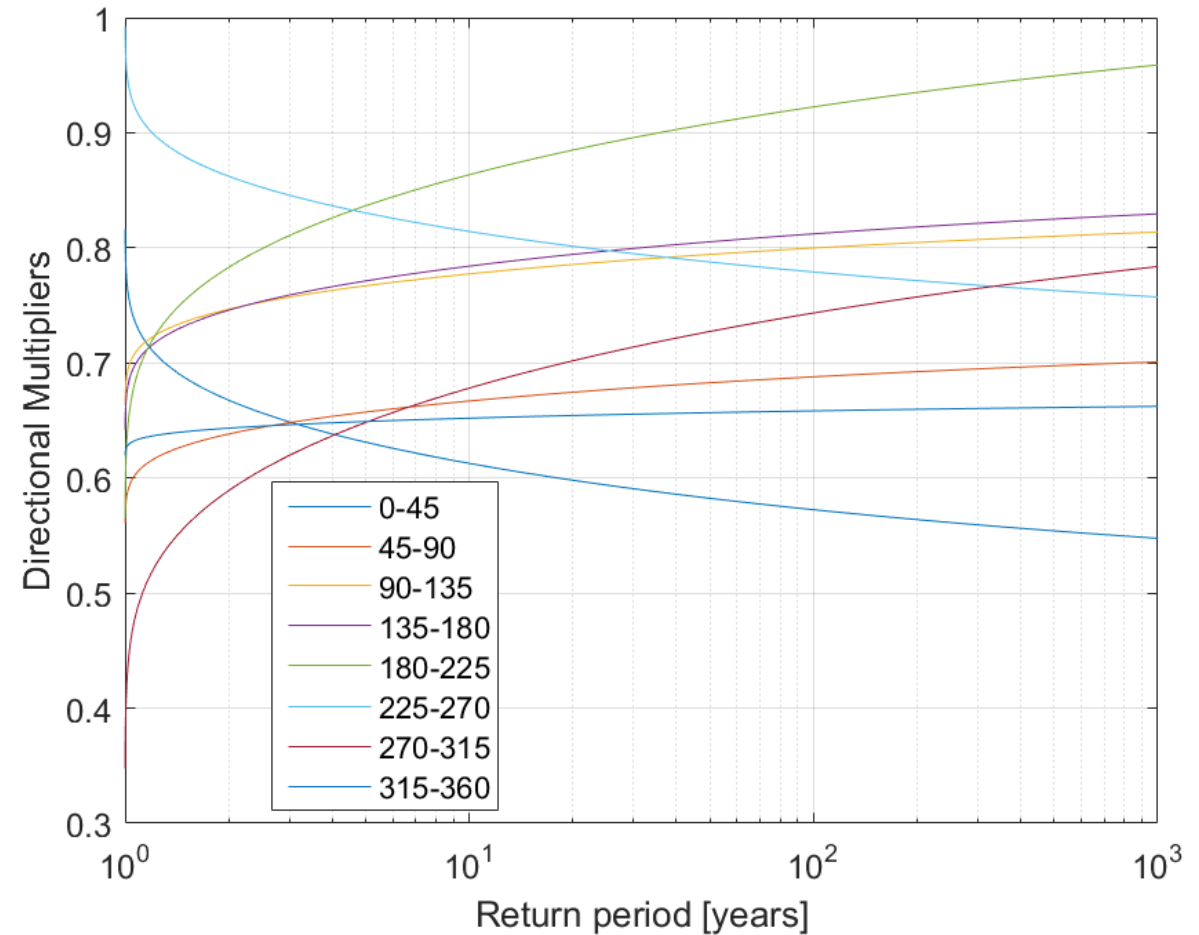
1. Divide directional return period curves by the all-direction curve (unconservative).
2. Distribution of target exceedance probability (Cook Method), (conservative)
3. Blend of 1 & 2 (Melbourne Method), applied in AS/NZS1170.2

Wind direction

Method 1

Final determination of
Md is subjective.

Method 2 & 3 explored
in assignment.



Storm type

- Many different types of windstorm can generate ‘design level’ wind gusts. Areas with more than one dominant wind type are known as ‘mixed wind climates’.
- Since they are different physical mechanisms they should be treated separately.
- Often difficult to identify origin of gusts.
- Mixed wind climates will have a multi-linear exceedance probability curve.

Storm type

Dealing with mixed wind climates

- Separate data into storm type bins
- Analyse as previously

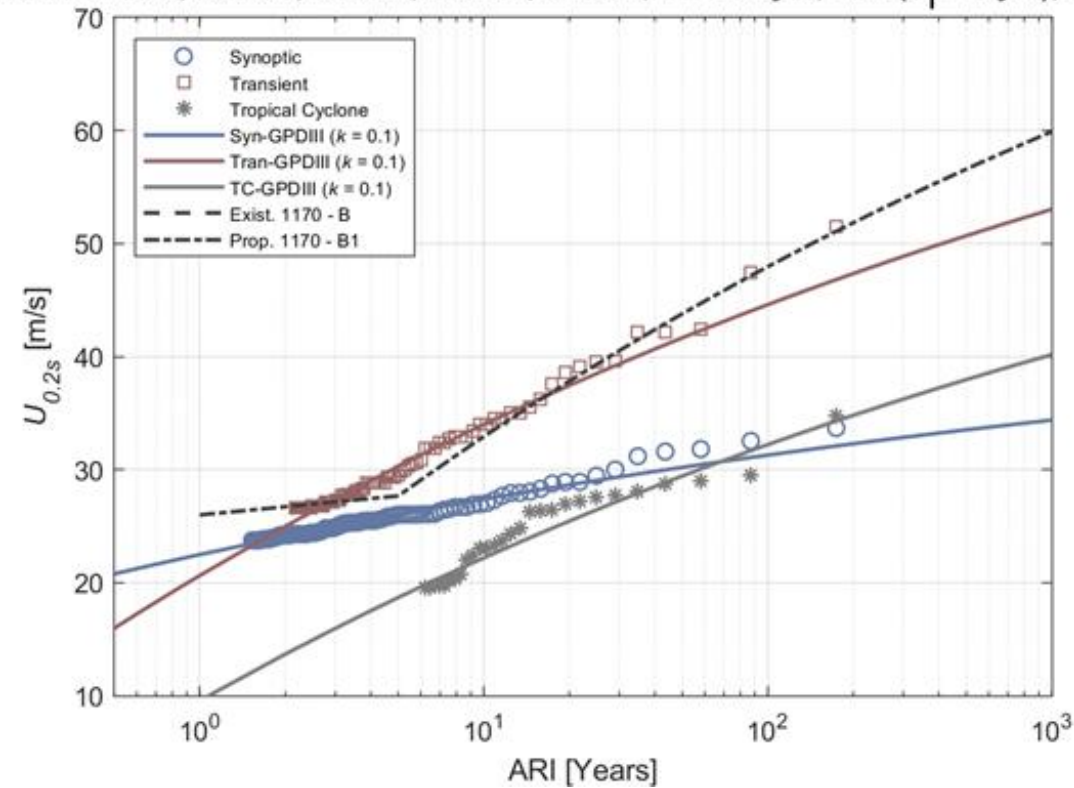
$$\left(1 - \frac{1}{R_{combined}}\right) = \prod_{i=1}^N \left(1 - \frac{1}{R_{Type_i}}\right)$$

Return period for gust with
data from all storm types

Return period for gust
from storm type i

Storm type

Station: 040004; 040842; 040223; 040211; 040082; $N = 174$ yrs; PoT ($u_T = 2$ yrs); L-Mom



Thunderstorm-generated gusts

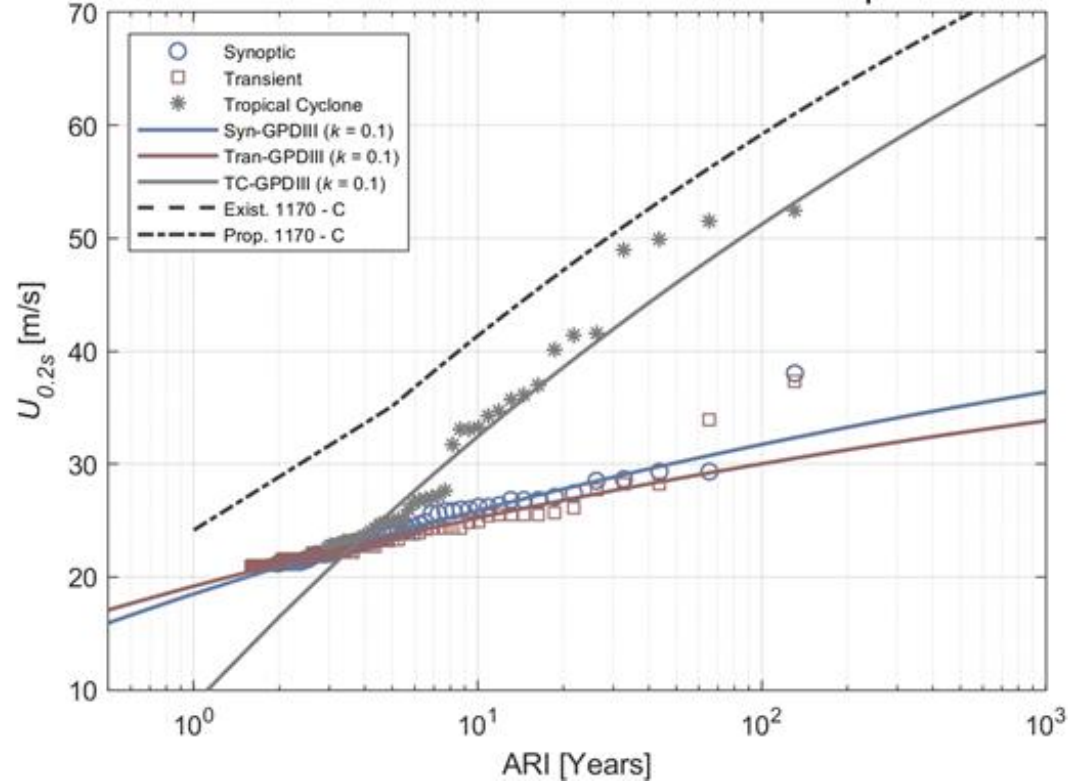
Tropical cyclone gusts

Synoptic gusts

Brisbane/Ipswich region

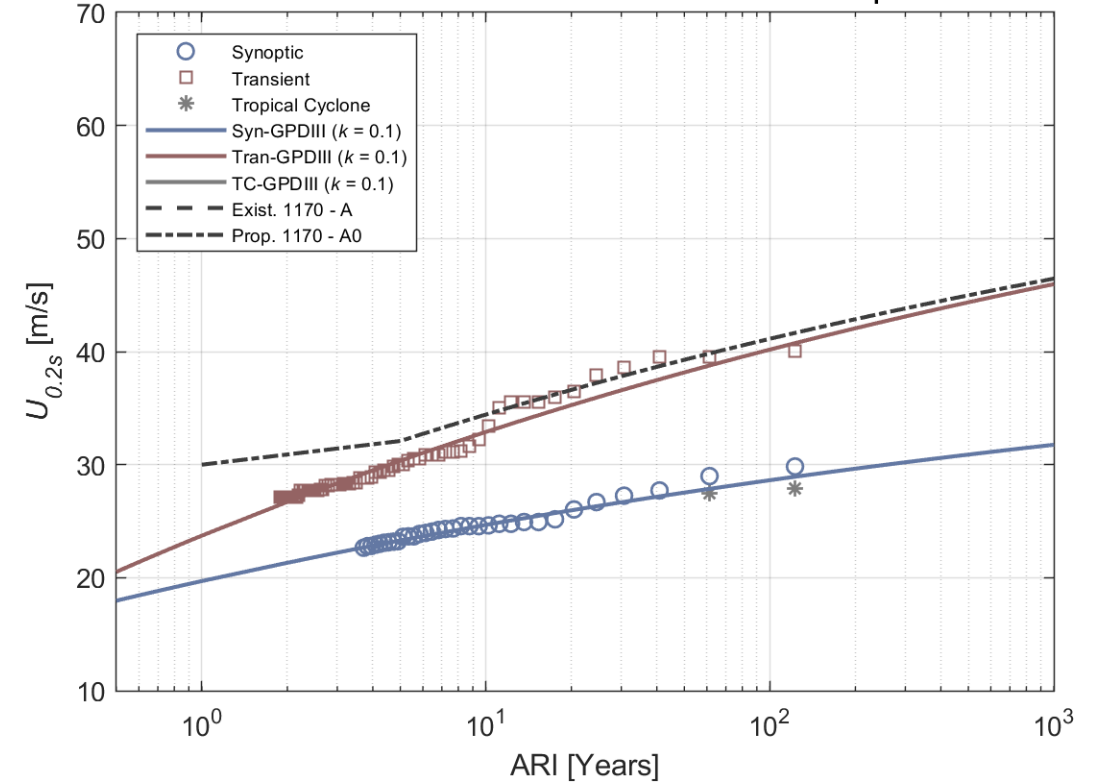
Storm type

Station: 031011; 031209; 031213; 032141; $N = 131$ yrs; PoT ($u_T = 2$ yrs); L-Mom fit



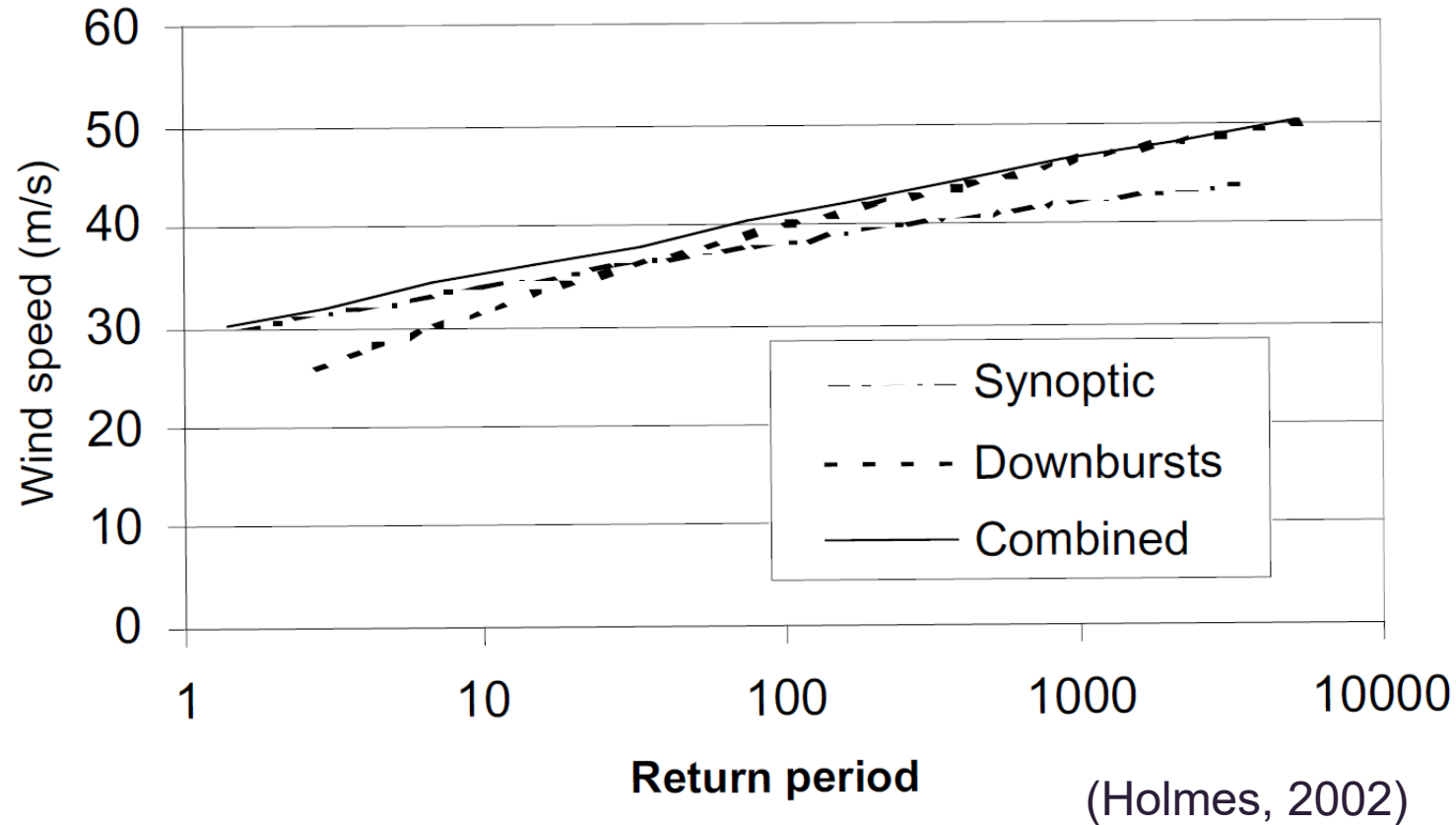
North Qld coast (north of Cairns)

Station: 043091; 043109; 044021; 045025; $N = 123$ yrs; PoT ($u_T = 2$ yrs); L-Mom fit



South-central Qld

Storm type



Extreme wind speeds in the Melbourne area
Essendon 1940-1971; Moorabbin 1972-92
Tullmarine 1970-97; Laverton 1946-95

Short data records

- Most places have insufficient history of recorded data (e.g. 20-50 years) to be confident in making predictions of long-term design wind speeds from a single recording station.
- Two primary methods used to address this:
 1. Compositing data (Superstations)
 2. Simulation techniques (tropical cyclones, thunderstorms)

Data compositing

- This technique involves grouping climatologically similar stations to artificially increase record lengths.
- Reduces sampling error introduced by short records.
- Local climate information is often lost.
- Assumption of independence of data from different composite stations – this is often difficult to ensure.
- Superstations are used regularly when developing wind loading codes and standards (e.g. Australia, USA)

Data compositing

Example: Composite wind data from weather stations in SE Qld (Brisbane Airport, Amberley AFB) and determine the 10, 100 and 1000 year return period wind gusts for each site and the superstation.

Assume: Gumbel is appropriate and we don't need to separate by storm type or direction.

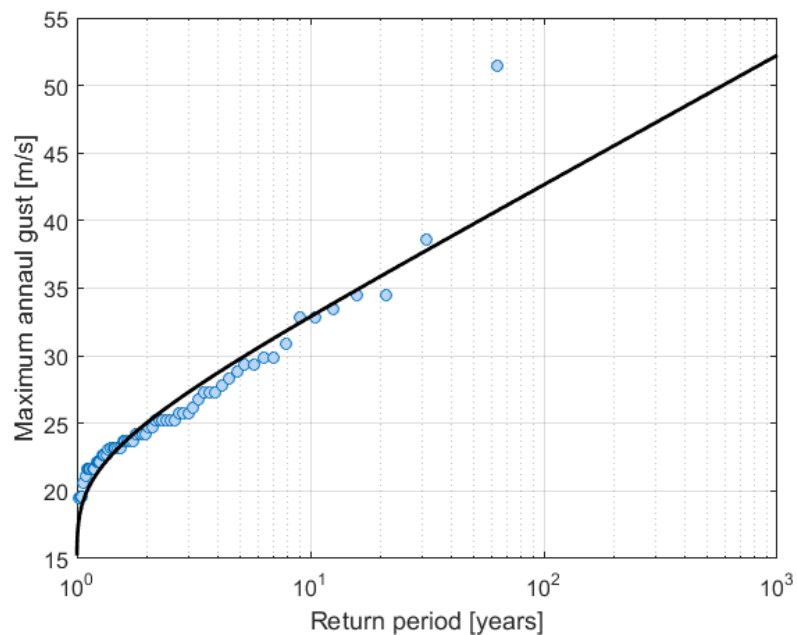
Stations:

Brisbane Airport (62 years)

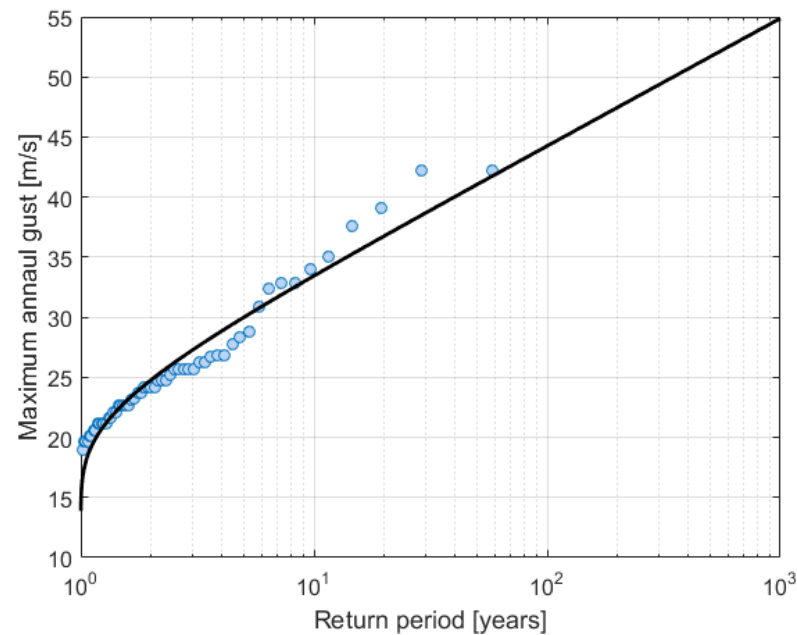
Amberley Air Force Base (58 years)

Total data record = 120 years

Data compositing

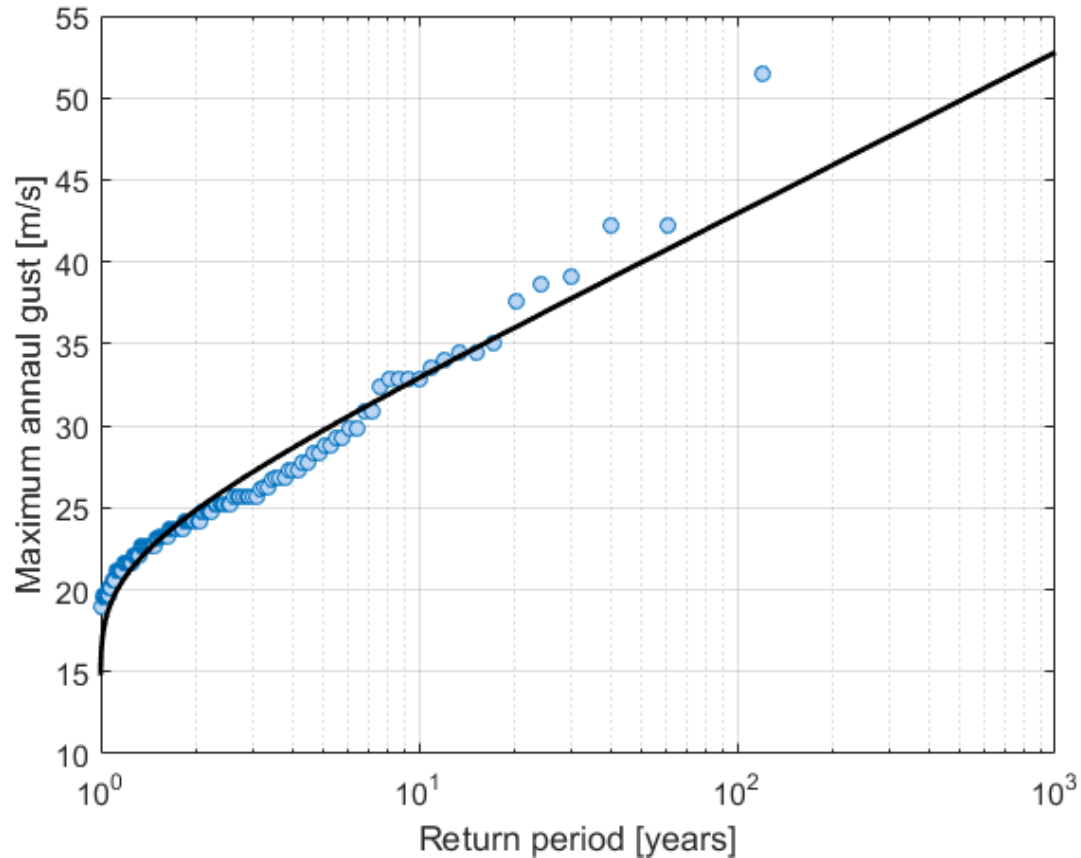


Brisbane Airport



Amberley Air Force Base

Data compositing



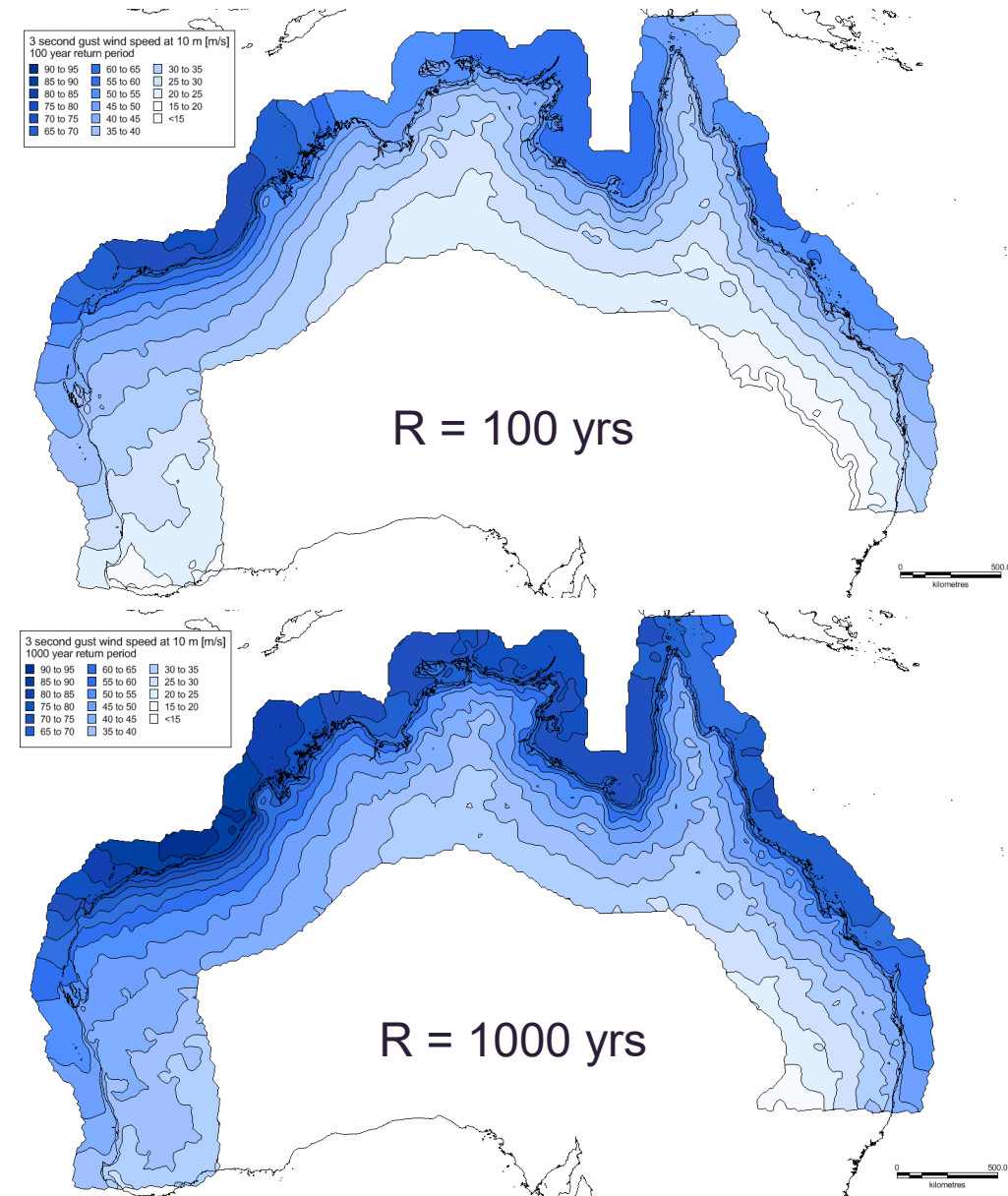
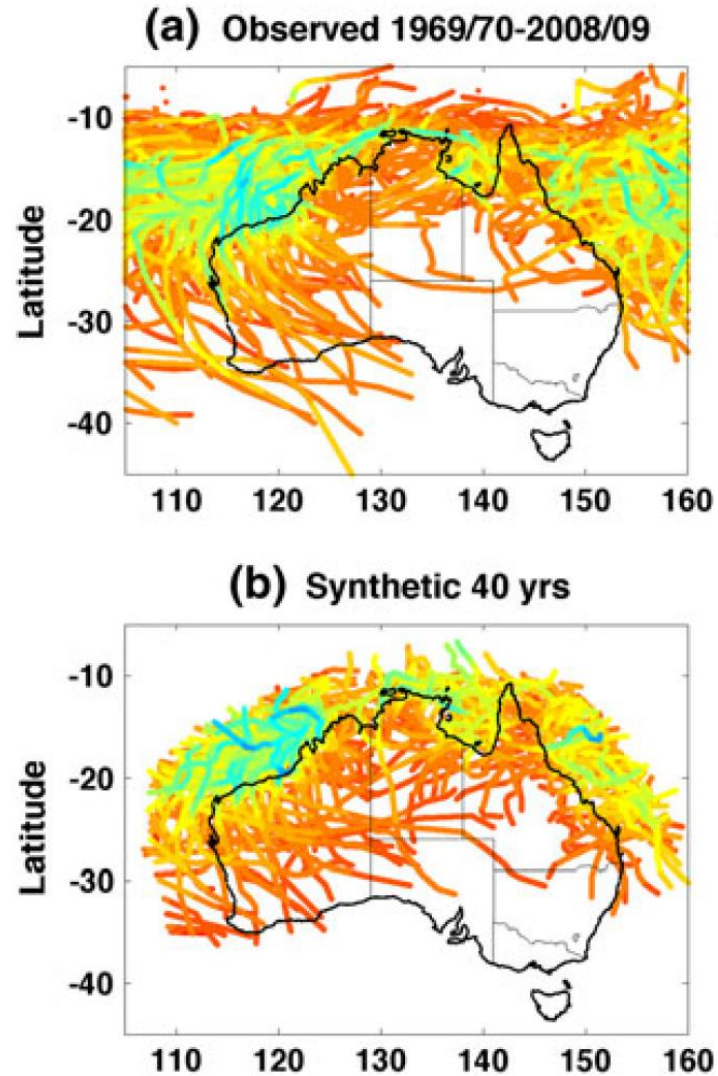
Combined Amberley and Brisbane AP

R [Years]	Brisbane AP [m/s]	Amberley [m/s]	Superstation [m/s]
10	32.9	33.5	32.9
100	42.7	44.3	42.9
1000	52.3	54.9	52.8

Simulation techniques

- Where the extreme wind climate is driven by rare events (e.g. tropical cyclones), simulation techniques can be used to artificially lengthen records.
- Can be purely statistical, or include storm physics.
- Complex models that draw on wind engineering, micrometeorology, meteorology, climatology research.
- Typical model includes sub-models that simulate:
 - Storm genesis
 - Storm behaviour (i.e. movement, intensification/decay)
 - Boundary layer structure
 - Interaction with underlying terrain/topography
 - Statistical aggregation

Simulation techniques



(Haigh et al. 2014)

Data correction/standardisation

- Most data is not perfect and requires some level of correction to a standardised reference (terrain category 2 (open terrain), flat, 10 m height, 3-second gust (0.2-second used in AS/NZS1170.2), 10-minute mean [WMO]).
- Typical corrections:
 - Instrumentation
 - Gust duration
 - Height
 - Topography/terrain
 - Station movement

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